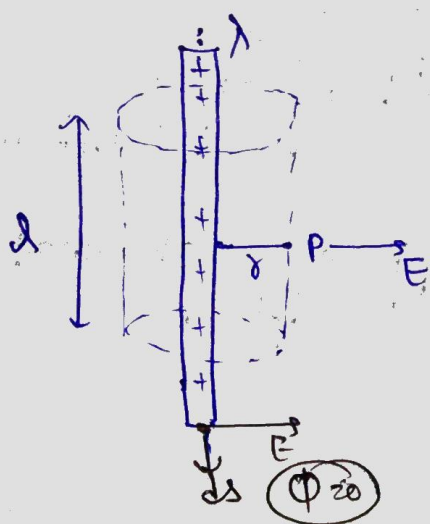


Application of Gauss's Law

- ① Electric field due to infinitely long straight wire.
- ② Uniformly charged infinite plane sheet.
- ③ Uniformly charged thin spherical shell.

① field Due to an infinitely long straight uniformly charged conductor.

Consider an infinitely long thin straight wire with uniform linear charge density λ .



To calculate the field, imagine a cylindrical Gaussian surface,
Since the field is everywhere radial. flux through the two ends of the cylindrical Gaussian surface is zero.
Imp.

* At the cylindrical part of the surface, E is normal to the surface at every point, and its magnitude is constant.